Matrix Sampling

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1 Notation

$||\cdot||$ represents the 2-norm.

$\lambda_{\max}(M)$ represents the maximum eigenvalue of the matrix M. Note: $\lambda_{\max}(M) = ||M||$, if M is PSD

M_{ij} represents the element in the i'th row, j'th column of the matrix M.

m_i represents the i'th row of the matrix M.

v_i represents the i'th entry in the vector v.

$\mathbb{E}$ represents expected value.

$\Pr(E)$ represents the probability that the random event E occurs.

$\mathbf{E}_{ij}$ 'represents the standard basis matrix, it has a one in the (i, j) position and zeros elsewhere'–Tropp.

2 Assumptions

1. $\max_{i\in[n]} ||a_i|| = 1$, i.e. $\forall i$, $||a_i|| \leq 1$. Any nonzero matrix can be rescaled s.t. $\max_{i\in[n]} ||a_i|| = 1$.
2. For all $i \in [n]$, $||a_i|| > 0$.
3. The row-magnitudes have a power-law distribution, i.e. if we assume without loss of generality that the rows of A are ordered by decreasing $||a_i||$, and satisfy $||a_i|| \leq ci^{-k}$, for $c \geq 0$ and $k \in (\frac{1}{3}, 1) \cup (1, \infty)$. A note on the $\frac{1}{3}$ term: this a relatively slowly-decreasing power-law distribution, which allows for many non-uniform matrices to fall within this requirement, while still resulting in strong concentration measures (to be proven).

3 The Sampling Technique

For any matrix $A \in \mathbb{R}^{n\times n}$, that matrix A can have its rows randomly sampled generating a random approximation to A, $\tilde{A}$, in the following manner:

$$\tilde{A} = DA = \sum_{j=1}^{n} \frac{Y_j}{\sqrt{||a_j||}} \mathbf{E}_{jj} A$$

Where $D \in \mathbb{R}^{n\times n}$ is a random diagonal matrix such that it's j'th diagonal entry contains the value $\frac{Y_j}{\sqrt{||a_j||}}$, and Y_j is a Bernoulli random variable, such that for all $j \in [n]$:

$$Y_j = \begin{cases} 1 & \text{w.p } ||a_j|| \\ 0 & \text{w.p. } 1 - ||a_j|| \end{cases}$$

Notice that $\mathbb{E}[Y_j] = ||a_j||$, and notice that is trivial to generate N independent and identically distributed copies of $\tilde{A}$. For clarity, the i'th such copy, $\tilde{A}_i$, can be defined as:

$$\tilde{A}_i = D_i A = \sum_{j=1}^{n} \frac{X_{ij}}{\sqrt{||a_j||}} \mathbf{E}_{jj} A$$

Where $D_i \in \mathbb{R}^{n \times n}$, is a random diagonal matrix D such that it's j'th diagonal entry contains the value $\frac{X_{ij}}{\sqrt{||a_j||}}$. And X_{ij} is a Bernoulli random variable, such that for all $i \in [N]$ and $j \in [n]$:

$$X_{ij} = \begin{cases} 1 & \text{w.p } ||a_j|| \\ 0 & \text{w.p. } 1 - ||a_j|| \end{cases}$$

Similarly, $\mathbb{E}[X_{ij}] = ||a_j||$.

Let $A^T A$ (the second moment of A), be defined to be the matrix $M \in \mathbb{R}^{n \times n}$ for simplicity's sake. M can be approximated by an average of N iid copies of the second moment of $\tilde{A}$:

$$\tilde{M} = \frac{1}{N} \sum_{i=1}^{N} \tilde{A}_i^T \tilde{A}_i = \frac{1}{N} \sum_{i=1}^{N} A^T D_i^2 A = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{n} \frac{X_{ij}}{||a_j||} a_j^T a_j$$

Notice that M is positive semi-definite, i.e. $\lambda_{\max}(M) = ||M||$. This is a crucial point, as the result to come will revolve around the two vectors $v, \tilde{v} \in \mathbb{R}^{n \times 1}$, where $||v|| = 1$ and $||\tilde{v}|| = 1$, and where v is such that $v^T M v = \lambda_{\max}(M)$ and $\tilde{v}^T \tilde{M} \tilde{v} = \lambda_{\max}(\tilde{M})$. Additionally, for the sake of notational concision, let the difference matrix, $\tilde{M} - M$ be defined to be $Z \in \mathbb{R}^{n \times n}$:

$$Z = \tilde{M} - M$$

One final definition of utiltiy is $Z_{ij} \in \mathbb{R}^{n \times n}$, which is defined as:

$$Z_{ij} = (\frac{X_{ij}}{||a_j||} - 1) \frac{a_j^T a_j}{N}$$

4 Theory

The goal of this section is to obtain that with high probability, $||A\tilde{w}|| \geq (1 - \epsilon)||Aw||$.

Lemma 4.1. $Z = \sum_{i=1}^{N} \sum_{j=1}^{n} Z_{ij}$

Proof.

$$\begin{aligned}
Z &= \tilde{M} - M \\
&= \frac{1}{N}\sum_{i=1}^{N} \tilde{A}_i^T \tilde{A}_i - A^T A \\
&= \frac{1}{N}\sum_{i=1}^{N} (\tilde{A}_i^T \tilde{A}_i - A^T A) \\
&= \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{n} (\frac{X_{ij}}{||a_j||} a_j^T a_j - a_j^T a_j) \\
&= \sum_{i=1}^{N}\sum_{j=1}^{n} (\frac{X_{ij}}{||a_j||} - 1)\frac{a_j^T a_j}{N} \\
&= \sum_{i=1}^{N}\sum_{j=1}^{n} Z_{ij}
\end{aligned}$$

□

Lemma 4.2. *For all $A, B \in \mathbb{R}^{n\times n}$, if A and B are Hermitian, then $\lambda_{\max}(A) - \lambda_{\max}(B) \leq \lambda_{\max}(A - B)$*

Proof.

$$\begin{aligned}
&\lambda_{\max}(B + (A - B)) \leq \lambda_{\max}(B) + \lambda_{\max}(A - B) \\
&\iff \lambda_{\max}(A) \leq \lambda_{\max}(B) + \lambda_{\max}(A - B) \\
\iff &\lambda_{\max}(A) - \lambda_{\max}(B) \leq \lambda_{\max}(A - B)
\end{aligned}$$

Notice that the first line holds due to Weyl's Inequality. □

Lemma 4.3. $\lambda_{\max}(A^T A) - \lambda_{\max}(\frac{1}{N}\sum_{i=1}^{N} \tilde{A}_i^T \tilde{A}_i) \leq ||Z||$

Proof.

$$\begin{aligned}
\lambda_{\max}(A^T A) - \lambda_{\max}(\frac{1}{N}\sum_{i=1}^{N} \tilde{A}_i^T \tilde{A}_i) &\leq \lambda_{\max}(A^T A - \frac{1}{N}\sum_{i=1}^{N} \tilde{A}_i^T \tilde{A}_i) && \textbf{Lemma 4.2} \\
&\leq ||A^T A - \frac{1}{N}\sum_{i=1}^{N} \tilde{A}_i^T \tilde{A}_i|| \\
&= ||\frac{1}{N}\sum_{i=1}^{N} \tilde{A}_i^T \tilde{A}_i - A^T A|| \\
&= ||Z||
\end{aligned}$$

□

Lemma 4.4. $\tilde{v}^T Z \tilde{v} \leq ||Z||$

Proof. Follows directly from definition of operator norm. □

Lemma 4.5. $\left|||A\tilde{v}||^2 - ||Av||^2\right| \leq 2||Z||$

Proof.

$$\begin{aligned}
\left|||A\tilde{v}||^2 - ||Av||^2\right| &= \left|\tilde{v}^T A^T A\tilde{v} - \lambda_{\max}(A^T A)\right| \\
&= \lambda_{\max}(A^T A) - \tilde{v}^T A^T A\tilde{v} \\
&= \lambda_{\max}(A^T A) - \lambda_{\max}(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i^T\tilde{A}_i) + \lambda_{\max}(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i^T\tilde{A}_i) - \tilde{v}^T A^T A\tilde{v} \\
&= \lambda_{\max}(A^T A) - \lambda_{\max}(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i^T\tilde{A}_i) + \tilde{v}^T(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i^T\tilde{A}_i)\tilde{v} - \tilde{v}^T A^T A\tilde{v} \\
&= \lambda_{\max}(A^T A) - \lambda_{\max}(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i^T\tilde{A}_i) + \tilde{v}^T(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i^T\tilde{A}_i - A^T A)\tilde{v} \\
&= \lambda_{\max}(A^T A) - \lambda_{\max}(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i^T\tilde{A}_i) + \tilde{v}^T Z\tilde{v} \\
&\leq ||Z|| + ||Z|| && \textbf{Lemmas 4.2, 4.4} \\
&= 2||Z||
\end{aligned}$$

□

Lemma 4.6. *For all $i \in [N]$, for all $j \in [n]$, $\mathbb{E}[Z_{ij}] = 0$*

Notice that this is the expected value of the summands of Z.

Proof.

$$\begin{aligned}
\mathbb{E}[Z_{ij}] &= \mathbb{E}[(\frac{X_{ij}}{||a_j||} - 1)\frac{a_j^T a_j}{N}] \\
&= (\frac{\mathbb{E}[X_{ij}]}{||a_j||} - 1)\frac{a_j^T a_j}{N} \\
&= (\frac{||a_j||}{||a_j||} - 1)\frac{a_j^T a_j}{N} \\
&= (1-1)\frac{a_j^T a_j}{N} \\
&= 0
\end{aligned}$$

□

Lemma 4.7. *For all $i \in [N]$, for all $j \in [n]$, $||Z_{ij}|| \leq \frac{1}{N}$*

Notice that this is an upper-bound on the summands of Z.

Proof.

$$
\begin{aligned}
||Z_{ij}|| &= ||(\frac{X_{ij}}{||a_j||} - 1)\frac{a_j^T a_j}{N}|| \\
&= \frac{1}{N}|\frac{X_{ij}}{||a_j||} - 1| \cdot ||a_j^T a_j|| \\
&= \frac{1}{N}|\frac{X_{ij}}{||a_j||} - 1| \cdot ||a_j||^2 && x \in \mathbb{R}^{n\times 1} \implies ||x^T x|| = ||x||^2 \\
&= \frac{1}{N}|X_{ij}||a_j|| - ||a_j||^2|\cdot \\
&= \begin{cases} (||a_j|| - ||a_j||^2)/N & \text{when, } X_{ij} = 1 \\ ||a_j||^2/N & \text{when, } X_{ij} = 0 \end{cases} \\
&\leq \begin{cases} 1/2N & \text{when, } X_{ij} = 1 \\ 1/N & \text{when, } X_{ij} = 0 \end{cases} && x \in [0,1] \implies x - x^2 \in [0, \frac{1}{2}] \\
&\leq \frac{1}{N}
\end{aligned}
$$

□

Lemma 4.8. *For all* $N > 0$, $\left|\left|\sum_{i=1}^N \sum_{j=1}^n \mathbb{E}[Z_{ij}^2]\right|\right| = \frac{1}{N}||\sum_{i=1}^n (||a_i|| - ||a_i||^2) a_i^T a_i||$

Notice that this is "the operator norm of the matrix variance of the sum"

Proof.

$$
\begin{aligned}
\left|\left|\sum_{i=1}^N \sum_{j=1}^n \mathbb{E}[Z_{ij}^2]\right|\right| &= \left|\left|\sum_{i=1}^N \sum_{j=1}^n \mathbb{E}\left[\left\{(\frac{X_{ij}}{||a_j||} - 1)\frac{a_j^T a_j}{N}\right\}^2\right]\right|\right| \\
&= \left|\left|\sum_{i=1}^N \sum_{j=1}^n \mathbb{E}\left[\frac{X_{ij}}{||a_j||^2} - 2\frac{X_{ij}}{||a_j||} + 1\right]\frac{a_j^T a_j a_j^T a_j}{N^2}\right|\right| \\
&= \left|\left|\sum_{i=1}^N \sum_{j=1}^n (\frac{\mathbb{E}[X_{ij}]}{||a_j||^2} - 2\frac{\mathbb{E}[X_{ij}]}{||a_j||} + 1)\frac{||a_j||^2 a_j^T a_j}{N^2}\right|\right| && \forall x \in \mathbb{R}^{1\times n}, xx^T = ||x||^2 \\
&= \left|\left|\sum_{i=1}^N \sum_{j=1}^n (\frac{1}{||a_j||} - 1)\frac{||a_j||^2 a_j^T a_j}{N^2}\right|\right| \\
&= \frac{1}{N}\left|\left|\frac{1}{N}\sum_{i=1}^N \sum_{j=1}^n (||a_j|| - ||a_j||^2) a_j^T a_j\right|\right| \\
&= \frac{1}{N}\left|\left|\sum_{i=1}^n (||a_i|| - ||a_i||^2) a_i^T a_i\right|\right|
\end{aligned}
$$

□

Lemma 4.9. *For all* $N > 0$, $||\sum_{i=1}^N \sum_{j=1}^n \mathbb{E}[Z_{ij}^2]|| \leq \frac{1}{4N}||Av||^2$

Notice that this is "the operator norm of the matrix variance of the sum"

Proof.

$$
\begin{aligned}
||\sum_{i=1}^{N}\sum_{j=1}^{n}\mathbb{E}[Z_{ij}^2]|| &= \frac{1}{N}||\sum_{i=1}^{n}(||a_i|| - ||a_i||^2)a_i^T a_i|| && \textbf{Lemmma 4.8}\\
&\leq \frac{1}{4N}||\sum_{i=1}^{n} a_i^T a_i|| && ||a_i|| - ||a_i||^2 \in [0, \frac{1}{4}]\\
&= \frac{1}{4N}||A^T A||\\
&= \frac{1}{4N}\lambda_{\max}(A^T A)\\
&= \frac{1}{4N}||Av||^2
\end{aligned}
$$

□

Lemma 4.10. *For all* $N, t \geq 0$, $\Pr(||Z|| \geq t/2) \leq \exp\left(\ln(2n) - \frac{Nt^2/8}{t/6+\nu}\right)$, *where* $\nu = ||Av||^2/4$

Proof. The first line of the following proof holds by Matrix Bernstein's Inequality (5.4.1 in Vershyn's HDP), and Lemmas 4.1, 4.6, 4.7, and 4.9

$$
\begin{aligned}
\Pr(||Z|| \geq t/2) &\leq 2n\exp(-\frac{t^2/8}{\frac{t}{6N} + \frac{1}{N}\nu})\\
&= 2n\exp(-\frac{Nt^2/8}{t/6+\nu})\\
&= \exp(\ln(2n) - \frac{Nt^2/8}{t/6+\nu})
\end{aligned}
$$

□

Lemma 4.11. *For all* $N, t \geq 0$, $\Pr\left(\left|||A\tilde{v}||^2 - ||Av||^2\right| \geq t\right) \leq \exp\left(\ln(2n) - \frac{Nt^2/8}{t/6+\nu}\right)$, *where* $\nu = ||Av||^2/4$

Proof.

$$
\begin{aligned}
\Pr(\left|||A\tilde{v}||^2 - ||Av||^2\right| \geq t) &\leq \Pr(2||Z|| \geq t) && \textbf{Lemma 4.5}\\
&\leq \Pr(||Z|| \geq t/2)\\
&\leq \exp(\ln(2n) - \frac{Nt^2/8}{t/6+\nu}) && \textbf{Lemma 4.10}
\end{aligned}
$$

□

Lemma 4.12. *For all* $N > 0$, $\epsilon \in [0,1]$, *with probability at least* $1 - \exp\left(\ln(2n) - \frac{N||Av||^2(2\epsilon-\epsilon^2)^2/4}{(2\epsilon-\epsilon^2)/3+1/2}\right)$, $||A\tilde{v}|| \in [(1-\epsilon)||Av||, ||Av||]$

Notice that if $||Av||$ and ϵ are constants, then the probability approaches 1 as n approaches infinity, so long as $N = \omega(\log n)$

Proof. By Lemma 4.11, with probability at least $1 - \exp\left(\ln(2n) - \frac{Nt^2/8}{t/6+\nu}\right)$,

$$\left|||A\tilde{v}||^2 - ||Av||^2\right| \leq t$$

Let $t = ||Av||^2(2\epsilon - \epsilon^2)$, then, with probability at least $1 - \exp\left(\ln(2n) - \frac{N||Av||^4(2\epsilon-\epsilon^2)^2/8}{||Av||^2(2\epsilon-\epsilon^2)/6+\nu}\right)\ldots$

$$\left|||A\tilde{v}||^2 - ||Av||^2\right| \leq ||Av||^2(2\epsilon - \epsilon^2)$$

Notice first that the probability can be simplified since $\nu = ||Av||^2/4$, thus the probability that $\left|||A\tilde{v}||^2 - ||Av||^2\right| \leq ||Av||^2(2\epsilon - \epsilon^2)$, holds is at least ...

$$1 - \exp\left(\ln(2n) - \frac{N||Av||^2(2\epsilon - \epsilon^2)^2/4}{(2\epsilon - \epsilon^2)/3 + 1/2}\right)$$

Note that by operator norm definition, $||A\tilde{v}||^2 \leq ||Av||^2$, so the lower bound is what is of significance, i.e.

$$\begin{aligned} -||Av||^2(2\epsilon - \epsilon^2) &\leq ||A\tilde{v}||^2 - ||Av||^2 \\ \iff (1 - 2\epsilon + \epsilon^2)||Av||^2 &\leq ||A\tilde{v}||^2 \\ \iff (1-\epsilon)^2||Av||^2 &\leq ||A\tilde{v}||^2 \\ \iff (1-\epsilon)||Av|| &\leq ||A\tilde{v}|| \end{aligned}$$

$\therefore$ With probability at least $1 - \exp\left(\ln(2n) - \frac{N||Av||^2(2\epsilon-\epsilon^2)^2/4}{(2\epsilon-\epsilon^2)/3+1/2}\right)$,

$$||A\tilde{v}|| \in [(1-\epsilon)||Av||, ||Av||]$$

□

Lemma 4.13. *For all* $N > 0$, $\frac{2}{||Av||}\sqrt{\frac{\ln(2n)}{N}} \leq \epsilon \leq 1$, *with probability at least* $1 - \exp\left(\ln(2n) - \frac{1}{4}N||Av||^2\epsilon^2\right)$, $||A\tilde{v}|| \geq (1-\epsilon)||Av||$

Proof. By Lemma 4.12 $||A\tilde{v}|| \in [(1-\epsilon)||Av||, ||Av||]$ holds with probability at least ...

$$\begin{aligned} 1 - \exp\left(\ln(2n) - \frac{N||Av||^2(2\epsilon - \epsilon^2)^2/4}{(2\epsilon - \epsilon^2)/3 + 1/2}\right) &\geq 1 - \exp\left(\ln(2n) - \frac{1}{4}N||Av||^2\epsilon^2(2-\epsilon)^2\right) && \forall x, (2x - x^2)/3 + 1/2 \leq 1 \\ &\geq 1 - \exp\left(\ln(2n) - \frac{1}{4}N||Av||^2\epsilon^2\right) && \forall x \in [0,1], x^2(2-x)^2 \geq x^2 \end{aligned}$$

□

Note that this is only a genuine probability when $1 - \exp(\ln(2n) - \frac{1}{4}N||Av||^2\epsilon^2) \geq 0$...

$$\begin{aligned}
1 - \exp(\ln(2n) - \frac{1}{4}N||Av||^2\epsilon^2) &\geq 0 \\
&\iff 1 \geq \exp(\ln(2n) - \frac{1}{4}N||Av||^2\epsilon^2) \\
&\iff 0 \geq \ln(2n) - \frac{1}{4}N||Av||^2\epsilon^2 \\
\iff \frac{1}{4}N||Av||^2\epsilon^2 &\geq \ln(2n) \\
&\iff \epsilon^2 \geq \frac{4\ln(2n)}{N||Av||^2} \\
&\iff \epsilon \geq \frac{2}{||Av||}\sqrt{\frac{\ln(2n)}{N}}
\end{aligned}$$

Lemma 4.14. *If* $||A\tilde{v}||^2 \geq (1-\epsilon)||Av||^2$, *then* $||M\tilde{v}|| \geq (1-\epsilon)||Mv||$

Proof.

$$\begin{aligned}
||M\tilde{v}|| &= ||\tilde{v}|| \cdot ||M\tilde{v}|| \\
&\geq |\langle \tilde{v}, M\tilde{v}\rangle| && \text{Cauchy Schwarz} \\
&= \tilde{v}^T M\tilde{v} \\
&= \tilde{v}^T A^T A\tilde{v} \\
&= ||A\tilde{v}||^2 \\
&\geq (1-\epsilon)||Av||^2 && \text{assumption} \\
&= (1-\epsilon)||Mv||
\end{aligned}$$

□

Lemma 4.15. *For all* $N > 0$, $\epsilon \in \left[\frac{2\ln(2n) \pm \sqrt{4\ln^2(2n) + 18\ln(2n)N||Av||^2}}{3N||Av||^2}, 1\right]$, *with probability at least* $1 - \exp\left(\ln(2n) - \frac{3N\epsilon^2||Av||^2}{4\epsilon+6}\right)$, $||M\tilde{v}|| \in [(1-\epsilon)||Mv||, ||Mv||]$

Notice that if $||Av||$ and ϵ are constants, then the probability approaches 1 as n approaches infinity, so long as $N = \omega(\log n)$

Proof. By Lemma 4.11, with probability at least $1 - \exp\left(\ln(2n) - \frac{Nt^2/8}{t/6+\nu}\right)$,

$$\left|||A\tilde{v}||^2 - ||Av||^2\right| \leq t$$

Let $t = \epsilon||Av||^2$, then, with probability at least $1 - \exp\left(\ln(2n) - \frac{N\epsilon^2||Av||^4/8}{\epsilon||Av||^2/6+\nu}\right)$...

$$\left|||A\tilde{v}||^2 - ||Av||^2\right| \leq \epsilon||Av||^2$$

Notice first that the probability can be simplified since $\nu = ||Av||^2/4$, thus the probability that $\left|||A\tilde{v}||^2 - ||Av||^2\right| \leq \epsilon||Av||^2$, holds is at least ...

$$1 - \exp\left(\ln(2n) - \frac{3N\epsilon^2||Av||^2}{4\epsilon+6}\right)$$

So, with probability at least $1 - \exp\left(\ln(2n) - \frac{3N\epsilon^2||Av||^2}{4\epsilon+6}\right)$,

$$\begin{aligned}
&\left|||A\tilde{v}||^2 - ||Av||^2\right| \leq \epsilon||Av||^2 \\
&\quad \iff -\epsilon||Av||^2 \leq ||A\tilde{v}||^2 - ||Av||^2 \leq \epsilon||Av||^2 \\
&\iff (1-\epsilon)||Av||^2 \leq ||A\tilde{v}||^2 \leq (1+\epsilon)||Av||^2 \\
&\iff (1-\epsilon)||Av||^2 \leq ||A\tilde{v}||^2 \leq ||Av||^2
\end{aligned}$$

$\therefore$ By Lemma 4.14, with probability at least $1 - \exp\left(\ln(2n) - \frac{3N\epsilon^2||Av||^2}{4\epsilon+6}\right)$,

$$||M\tilde{v}|| \in [(1-\epsilon)||Mv||, ||Mv||]$$

Note that this is only a valid probability when $1 - \exp\left(\ln(2n) - \frac{3N\epsilon^2||Av||^2}{4\epsilon+6}\right) \geq 0$:

$$\begin{aligned}
1 - \exp\left(\ln(2n) - \frac{3N\epsilon^2||Av||^2}{4\epsilon+6}\right) &\geq 0 \\
1 &\geq \exp\left(\ln(2n) - \frac{3N\epsilon^2||Av||^2}{4\epsilon+6}\right) \\
0 &\geq \ln(2n) - \frac{3N\epsilon^2||Av||^2}{4\epsilon+6} \\
\frac{3N\epsilon^2||Av||^2}{4\epsilon+6} &\geq \ln(2n) \\
3N\epsilon^2||Av||^2 &\geq 4\ln(2n)\epsilon + 6\ln(2n) \\
3N||Av||^2\epsilon^2 - 4\ln(2n)\epsilon - 6\ln(2n) &\geq 0 \\
\epsilon &\geq \frac{4\ln(2n) + \sqrt{16\ln^2(2n) + 72\ln(2n)N||Av||^2}}{6N||Av||^2} \\
\epsilon &\geq \frac{2\ln(2n) + \sqrt{4\ln^2(2n) + 18\ln(2n)N||Av||^2}}{3N||Av||^2}
\end{aligned}$$

□

4.1 The Algorithm

Theoretically, one of the most efficient ways to compute $\tilde{M}$ is to compute n outer-products in parallel, along with a Bernoulli distribution. This looks like $\tilde{M} = \sum_{i=1}^{n} \frac{1}{N||a_i||} a_i^T a_i \cdot \text{Binomial}(N, ||a_i||)$. Notice that the work can theoretically be distributed to one thread per row, followed by a summation of n scaled outer-products. See the following Lemma which justifies $\tilde{M}$ as the sum of Binomial random variable scaled outer-products.

Lemma 4.16. *For all $N > 0$, $\tilde{M} = \sum_{i=1}^{n} \frac{1}{N||a_i||} a_i^T a_i \cdot Binomial(N, ||a_i||)$*

Proof.

$$\begin{aligned}
\tilde{M} &= \frac{1}{N}\sum_{j=1}^{N}\sum_{i=1}^{n}\frac{X_{ji}}{||a_i||}a_i^T a_i \\
&= \sum_{i=1}^{n}\frac{1}{N\cdot||a_i||}a_i^T a_i \sum_{j=1}^{N} X_{ji} \\
&= \sum_{i=1}^{n}\frac{1}{N\cdot||a_i||}a_i^T a_i \cdot \text{Binomial}(N, ||a_i||)
\end{aligned}$$

□

4.2 Why The Second Moment Must be Used to Generate $\tilde{v}$

Lemma 4.17. $\lambda_{\max}(\tilde{M}) \geq \lambda_{\max}\left(\left(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i\right)^T\left(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i\right)\right)$

Note that this implies that an average of the first moment is not sufficient in reconstructing an average of the second moment, akin to how an average of squares is not necessarily equivalent to the square of averages. In other words, the average $\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i$ is insufficient for generating $\tilde{v}$, and $\tilde{v}$ must be generated from $\tilde{M}$.

Proof. Let $B = \left(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i\right)^T\left(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i\right)$

$$\begin{aligned}
\tilde{M} - B &= \tilde{M} - B - B + B \\
&= \frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i^T\tilde{A}_i - \left(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i\right)^T\left(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i\right) - \left(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i\right)^T\left(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i\right) + \left(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i\right)^T\left(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i\right) \\
&= \frac{1}{N}\sum_{i=1}^{N}\left\{\tilde{A}_i^T\tilde{A}_i - \tilde{A}_i^T\left(\frac{1}{N}\sum_{j=1}^{N}\tilde{A}_j\right) - \left(\frac{1}{N}\sum_{i=j}^{N}\tilde{A}_j\right)^T\tilde{A}_i + \left(\frac{1}{N}\sum_{i=j}^{N}\tilde{A}_j\right)^T\left(\frac{1}{N}\sum_{j=1}^{N}\tilde{A}_j\right)\right\} \\
&= \frac{1}{N}\sum_{i=1}^{N}\left(\tilde{A}_i - \frac{1}{N}\sum_{j=1}^{N}\tilde{A}_j\right)^T\left(\tilde{A}_i - \frac{1}{N}\sum_{j=1}^{N}\tilde{A}_j\right) \\
&\geq 0
\end{aligned}$$

Note that the final line from the above algebra holds due to the fact that the sum of PSD matrices is itself a PSD matrix.

Since $\tilde{M} - \left(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i\right)^T\left(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i\right) \geq 0$, then

$$\lambda_{\max}(\tilde{M}) \geq \lambda_{\max}\left(\left(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i\right)^T\left(\frac{1}{N}\sum_{i=1}^{N}\tilde{A}_i\right)\right)$$

□

4.3 Sampling Amount

The goal of this section is to determine the expected amount of sampling that occurs.

Let…

- $I(\cdot)$ be the indicator function
- $nnz(\cdot)$ be a function which takes a matrix as input and outputs the number of nonzero entries in that matrix
- $N \in \mathbb{Z}^+$ be the number of binomial trials
- $A \in \mathbb{R}^{n \times m}$ be the matrix to approximate the second moment of
- $a_i \in \mathbb{R}^m$ be the i'th row of A
- η_i be the number of nonzero entries in a_i
- S_i be a set containing the indices of the nonzero entries of a_i. Note: $|S_i| = \eta_i$.
- For all $\ell \in [N]$, $i \in [n]$, let $X_{\ell i} = \begin{cases} 1 & \text{w.p. } ||a_i|| \\ 0 & \text{w.p. } 1 - ||a_i|| \end{cases}$
- $\tilde{M} \in \mathbb{R}^{n \times n}$ be the approximation of $A^T A$:

$$\tilde{M} = \frac{1}{N} \sum_{i=1}^{N} \tilde{A}_i^T \tilde{A}_i = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{n} \frac{X_{ij}}{||a_j||} a_j^T a_j$$

To determine the expected number of nonzeros in the matrix $\tilde{M}$, the following function is useful. The function $f(i, j, k)$ indicates wether the (j, k)'th entry of $\tilde{M}$ is nonzero when using the first i outer-products to generate the approximation. More specifically, for all $i, j, k \in [n]$:

$$f(i, j, k) = \begin{cases} 1 & \text{if the matrix } \sum_{r=1}^{i} I\left(\sum_{\ell=1}^{N} X_{\ell r} > 0\right) a_r^T a_r \text{ has a a nonzero entry in the } (j, k)\text{'th position} \\ 0 & \text{otherwise} \end{cases}$$

Lemma 4.18. $\mathbb{E}[nnz(\tilde{M})] = \sum_{j,k \in [n]} \Pr(f(n, j, k) = 1)$

Proof.

$$\begin{aligned} \mathbb{E}[nnz(\tilde{M})] &= \mathbb{E}\left[\sum_{j,k \in [n]} f(n, j, k)\right] \\ &= \sum_{j,k \in [n]} \mathbb{E}\left[f(n, j, k)\right] \\ &= \sum_{j,k \in [n]} \Pr(f(n, j, k) = 1) \end{aligned}$$

□

Notice that the function f can be concisely defined recursively:

$$f(i, j, k) = \begin{cases} 0 & \text{if } i = 0 \\ \max\left\{ f(i-1, j, k), I(j \in S_i) \cdot I(k \in S_i) \cdot I\left(\sum_{\ell=1}^{N} X_{\ell i} > 0\right) \right\} & \text{otherwise} \end{cases}$$

Similarly, $\Pr(f(i,j,k)=1)$ can also be defined recursively:

$$\Pr(f(i,j,k)=1) = \begin{cases} 0 & \text{if } i = 0 \\ \alpha + \beta - \alpha \cdot \beta & \text{otherwise} \end{cases}$$

where $\alpha = \Pr(f(r-1,j,k)=1)$ and $\beta = I(j \in S_i) \cdot I(k \in S_i) \cdot \Pr\left(\sum_{\ell=1}^{N} X_{\ell i} > 0\right)$.

4.4 Sampling Amount (OLD)

Let W be the random variable, that is the total number of nonzero rows of N iid copies of $\tilde{A}_i$, i.e. $W = \sum_{i=1}^{N} \sum_{j=1}^{n} X_{ij}$.

Lemma 4.19. $\mathbb{E}[W] = \sum_{i=1}^{N} \sum_{j=1}^{n} ||a_j||$.

Proof.

$$\begin{aligned} \mathbb{E}[W] &= \sum_{i=1}^{N} \sum_{j=1}^{n} \mathbb{E}[X_{ij}] && \text{LOE} \\ &= \sum_{i=1}^{N} \sum_{j=1}^{n} ||a_j|| \end{aligned}$$

□

Lemma 4.20. $\mathbb{E}[W] \leq \begin{cases} cN(1 + \frac{n^{-k+1}-1}{-k+1}) & k \neq 1 \\ cN(1+\ln(n)) & k = 1 \end{cases}$, *where* $c > 0$

Proof.

$$\begin{aligned} \mathbb{E}[W] &= \sum_{i=1}^{N} \sum_{j=1}^{n} ||a_j|| && \textbf{Lemma 4.19} \\ &\leq c \sum_{i=1}^{N} \sum_{j=1}^{n} j^{-k} && \text{WLOG } ||a_j|| \leq cj^{-k} \\ &= cN \sum_{j=1}^{n} j^{-k} \\ &\leq cN(1 + \int_{x=1}^{n} x^{-k} dx) \\ &= \begin{cases} cN(1 + \frac{n^{-k+1}-1}{-k+1}) & k \neq 1 \\ cN(1+\ln(n)) & k = 1 \end{cases} \end{aligned}$$

□

Lemma 4.21. $\mathbb{E}[W] = \begin{cases} O(Nn^{1-k}) & \textit{when } k \in (0,1) \\ O(N \log n) & \textit{when } k = 1 \\ O(N) & \textit{when } k \in (1,\infty) \end{cases}$

Proof.

$$\mathbb{E}[W] \leq \begin{cases} cN(1+\frac{n^{-k+1}-1}{-k+1}) & k \neq 1 \\ cN(1+\ln(n)) & k = 1 \end{cases} \qquad \textbf{Lemma 4.20}$$

$$= \begin{cases} O(Nn^{1-k}) & \text{when } k \in (0,1) \\ O(N\log n) & \text{when } k = 1 \\ O(N - \frac{N}{n^{k-1}}) & \text{when } k \in (1,\infty) \end{cases}$$

$$= \begin{cases} O(Nn^{1-k}) & \text{when } k \in (0,1) \\ O(N\log n) & \text{when } k = 1 \\ O(N) & \text{when } k \in (1,\infty) \end{cases}$$

□

4.5 Extra Stuff

This section involves some proofs which are aimed toward gaining more understanding of the implications of the found Matrix Bernstein bounds.

Lemma 4.22. *If $A \in \mathbb{R}^{1\times 1}$ $\lambda_{\max}(A) = \alpha$, then $\lambda_{\max}(A^TA) = \alpha^2$*

Proof.

$$\lambda_{\max}(A^TA) = \lambda_{\max}\left([\alpha]^T[\alpha]\right) \qquad \lambda_{\max}(A) = \alpha \implies A = [\alpha]$$
$$= \lambda_{\max}\left([\alpha^2]\right)$$
$$= \alpha^2$$

□